SS3 to opal: North Pacific swordfish

Bridging the 2023 ISC two-sex, quarterly assessment to a single-sex, annual Stock Synthesis model, and from it to opal

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1 Overview

The 2023 ISC benchmark assessment of North Pacific swordfish (*Xiphias gladius*) [1] uses Stock Synthesis [SS3 3.30.20; R. D. Methot and C. R. Wetzel [2]]. The model has two sexes, which differ in growth and natural mortality, and four seasons (quarters) per year with annual ages. Spawning is in the second quarter (month 5), recruits settle in month 7, and catch, indices and length compositions are entered by quarter. Two fleets have blocks of time-varying selectivity, and one index measures recruitment directly. opal has one sex. Its SS3 converter handles one season per year. Within a year, opal removes catch at the start of each season, and it has no recruitment index. The base case therefore cannot be converted directly.

This case study bridges in two stages:

1. **Within SS3** (Section 3), from the base case to a single-sex, annual SS3 model (B7) whose structure opal can represent. Each step makes one change, so its effect is measured on its own.
2. **SS3 to opal** (Section 4), converting B7 with the shared converter in `shared/ss3-to-opal/`. The conversion is checked and SS3's dynamics are replayed with opal's catch timing (Section 5). opal is then fitted (Section 6).

Section 7 lists where opal falls short of SS3 or differs from it, and what opal would need to model the base case itself. The case study documents these differences; it does not change opal.

In brief:

- The base case re-runs exactly with SS3 3.30.22.1 (NLL 289.3000, SSB_{2021} 35,778 t, SSB_{MSY} 16,388 t, as in the report's Table S2).
- **Single sex (B1)** is the largest change in the bridge. It applies female growth and female M to all fish. Females are larger and, at older ages, have much lower M than males (0.22 against 0.36). SSB_0 falls from 99,808 t to 78,149 t, and terminal depletion from 0.358 to 0.258. With sex-averaged M and growth instead (sensitivity B1s), depletion is 0.389, close to the base case.
- **The other steps** each move terminal depletion by at most 0.021. The blocks become fleets exactly. Merging the quarters requires spawning at the start of the year, and the B7 model has depletion 0.292.
- **The conversion of B7 is exact** where opal and SS3 share structure. This covers catch, indices, compositions, biology, the unfished equilibrium and selectivity-at-length. Replaying SS3′s numbers-at-age with SS3′s catch equation and catch weights reproduces SS3′s spawning biomass to 4.0e-06.
- **opal at SS3′s values (O0)** differs from SS3 through catch timing (up to 18% in spawning biomass from removing the year's catch at its start) and through its initial equilibrium (4.2% in 1975).
- **opal fitted with SS3′s selectivity (O1)** converges and estimates terminal depletion 0.336 (SS3 B7: 0.292). opal's own selectivity curves, started at SS3′s, are fitted in O2. Eight of their parameters end on a bound or cannot be determined. Fixed at their O2 values (O2b), the fit converges with the same objective and estimates depletion 0.305.

2 The 2023 assessment

2.1 Stock, fisheries and data

The stock area is the North Pacific north of the equator in the WCPFC Convention Area and north of 10°N in the IATTC Convention Area. The model has one area and models the western-central and eastern North Pacific with fleets as areas. It runs from 1975 to 2021 with four quarterly seasons per year. Ages are annual (0-15, plus group at 15), with 5 cm population length bins from 10 to 260 cm.

Table 1: Fleets of the base case and of B7 as opal fisheries. Units: catch (or index) in tonnes (1) or numbers (2). Compositions: fitted length compositions (lambda 0.5 in SS3).

opal fishery	SS3 fleet	Name	Units	Compositions
1	1	F1_JPN_WCNPO_OSDWCOLL_late_Area1	2	yes
2	2	F2_TWN_WCNPO_DWLL_late	1	yes
3	3	F3_US_WCNPO_LL_shallow_late	2	yes
4	4	F4_IATTC	1	yes
5	5	F5_JPN_EPO_OSDWLL	2	
6	6	F6_JPN_WCNPO_OSDWLL_early_Area1	2	yes
7	7	F7_JPN_WCNPO_CODF	1	yes
8	8	F8_US_WCNPO_LL_shallow_early	2	yes
9	9	F9_US_WCNPO_LL_deep	2	
10	10	F10_JPN_WCNPO_OSDF	1	
11	11	F11_JPN_WCNPO_Other_early	1	
12	12	F12_JPN_WCNPO_Other_late	1	
13	13	F13_TWN_WCNPO_DWLL_early	1	
14	14	F14_TWN_WCNPO_Other	1	
15	15	F15_US_WCNPO_GN	1	
16	16	F16_US_WCNPO_Other	1	
17	17	F17_JPN_WCNPO_OSDWLL_early_Area2	2	
18	18	F18_JPN_WCNPO_OSDWLL_late_Area2	2	
19	19	F19_WCPFC	1	
20	28	F1_JPN_WCNPO_OSDWCOLL_late_Area1_1994_97	2	yes
21	29	F12_JPN_WCNPO_Other_late_1994_97	1	
22	31	F2_TWN_WCNPO_DWLL_late_2004_09	1	yes
23	32	F15_US_WCNPO_GN_2004_09	1	
24	33	F16_US_WCNPO_Other_2004_09	1	
25	34	F19_WCPFC_2004_09	1	
26	20	S1_JPN_WCNPO_OSDWLL_early_Area1	2	
27	21	S2_JPN_WCNPO_OSDWCOLL_late_Area1	2	
28	22	S3_JPN_WCNPO_OSDWLL_early_Area2	2	
29	23	S4_JPN_WCNPO_OSDWLL_late_Area2	2	
30	25	S6_US_WCNPO_LL_deep	2	
31	26	S7_US_WCNPO_LL_shallow_early	2	
32	30	S2_JPN_WCNPO_OSDWCOLL_late_Area1_1994_97	2	

The base case has 19 fishing fleets. Japanese and US longline catches are in thousands of fish and all others in tonnes. It has eight abundance indices, of which S5 (Chinese Taipei longline) and S8 (US shallow-set longline, late period) have lambda 0. S6, the US deep-set longline index, is a **recruitment index**: SS3 compares it with the year's age-0 recruitment (units code 33), so its selectivity plays no part. Length compositions (sex-

combined) are entered by quarter. Those of eight fleets are fitted with lambda 0.5. F5′s have lambda 0, and every F9 observation is excluded (negative year).

2.2 Biology and structure

Table 2: Structure of the base case, from the model files.

Item	Base case
SS3 version	3.30.20.00
Sexes	2; sex ratio at birth 1:1; SSB is female spawning biomass
Seasons	4 quarters, 2 subseasons
Ages	0-15 (annual)
Growth	von Bertalanffy, L1 at age 0.5, L2 at age 15; females 80.1 / 226.3 cm, K 0.246; males 83.4 / 206.4 cm, K 0.271; CV 0.1 (fixed)
Natural mortality	age-specific, fixed; females 0.42 at age 0 to 0.22 from age 4; males 0.40 to 0.36
Maturity and fecundity	length logistic (L50 143.7 cm); fecundity = weight
Spawning and settlement	spawning month 5 (quarter 2); settlement month 7 at age 0
Stock-recruitment	Beverton-Holt, h = 0.9, sigma_R = 0.42 (fixed)
Recruitment deviations	early 1965-1984, main 1985-2021, bias ramp
Fishing mortality	hybrid (F method 3); F-ballpark off
Selectivity	length-based double normal (F1, F3, F6, F7, F8) or logistic (F2, F4); other fleets mirror; blocks on F1 (1994-97) and F2 (2004-09)
Initial equilibrium	initial F for F6 (initial-catch lambda 0)
Catchability	floated q for every index
Data weighting	length compositions lambda 0.5; S5, S8 and F5 compositions lambda 0

The sexes share selectivity (no sex offsets), weight-length and maturity, and the compositions are sex-combined. The two sexes therefore differ only in growth and M. Males are smaller (L2 206.4 against 226.3 cm). From age 4, males have M of 0.36, against 0.22 for females. The report notes that the lack of sex-specific size data is one of the two main sources of uncertainty in the assessment. Section 2.6 of the report describes the model as single sex, but the model files have two sexes.

3 The SS3 bridge

`scripts/01-build-ss3-bridge.r` builds each step and estimates it with SS3 3.30.22.1, Hessian included. Each step makes one change:

- **B0**: as supplied, re-run with 3.30.22.1 from the supplied par file.

- **B1**: single sex (`Nsexes = -1`, so spawning biomass is that of the single sex times the female fraction 0.5). Female growth, M and weight-length are used, and the sex-combined compositions keep one copy of their vector.
- **B1s** (sensitivity, built from B0): as B1, with sex-averaged biology. M at age is the rate that takes the unfished numbers of both sexes from one age to the next. L1, L2 and K are fitted to the mean length-at-age of the two sexes, weighted by their unfished numbers.
- **B2**: the selectivity blocks become fleets, together with every fleet that mirrors a blocked fleet. The 1994-97 fleets are F1, F12 and the index S2 (fleets 28-30). The 2004-09 fleets are F2, F15, F16 and F19 (fleets 31-34). F13 and F14 mirror F2 and F1 but have no catch in the block years. S5 has lambda 0 and is not split.
- **B3**: spawning in month 1 (season 1) instead of month 5.
- **B4**: one season, with the catches summed over the quarters.
- **B5**: one length composition per fleet and year, combining the quarters (sample sizes summed, proportions weighted by sample size).
- **B6**: recruits settle in month 1 at age 0 instead of month 7.
- **B7**: indices observed in month 1.

Table 3: The SS3 bridge. NLL is SS3's total; B5 changes its constant (see text). F2021 is the average F over ages 1-10. All steps converged (maximum gradient below 2e-3) with a positive-definite Hessian.

Step	Est. par.	NLL	Max. grad.	SSB0	SSB2021	SSB2021/ SSB0	F2021	SSB MSY
B0 base case	86	289.30	5.8e-05	99,808	35,778	0.358	0.087	16,388
B1 single sex	86	293.40	2.4e-05	78,149	20,197	0.258	0.118	14,996
B1s sex-averaged biology	86	292.09	9.7e-05	66,937	26,008	0.389	0.089	11,818
B2 blocks as fleets	86	292.65	5.2e-06	78,558	20,308	0.259	0.118	15,103
B3 spawning month 1	86	292.59	2.0e-05	83,206	23,245	0.279	0.118	16,975
B4 annual	86	291.69	1.6e-05	82,574	22,717	0.275	0.120	16,507
B5 annual compositions	86	42.79	1.9e-03	82,574	22,717	0.275	0.120	16,507
B6 settlement month 1	86	36.49	4.6e-05	81,740	23,833	0.292	0.134	15,949
B7 index month 1	86	36.49	4.6e-05	81,740	23,833	0.292	0.134	15,949

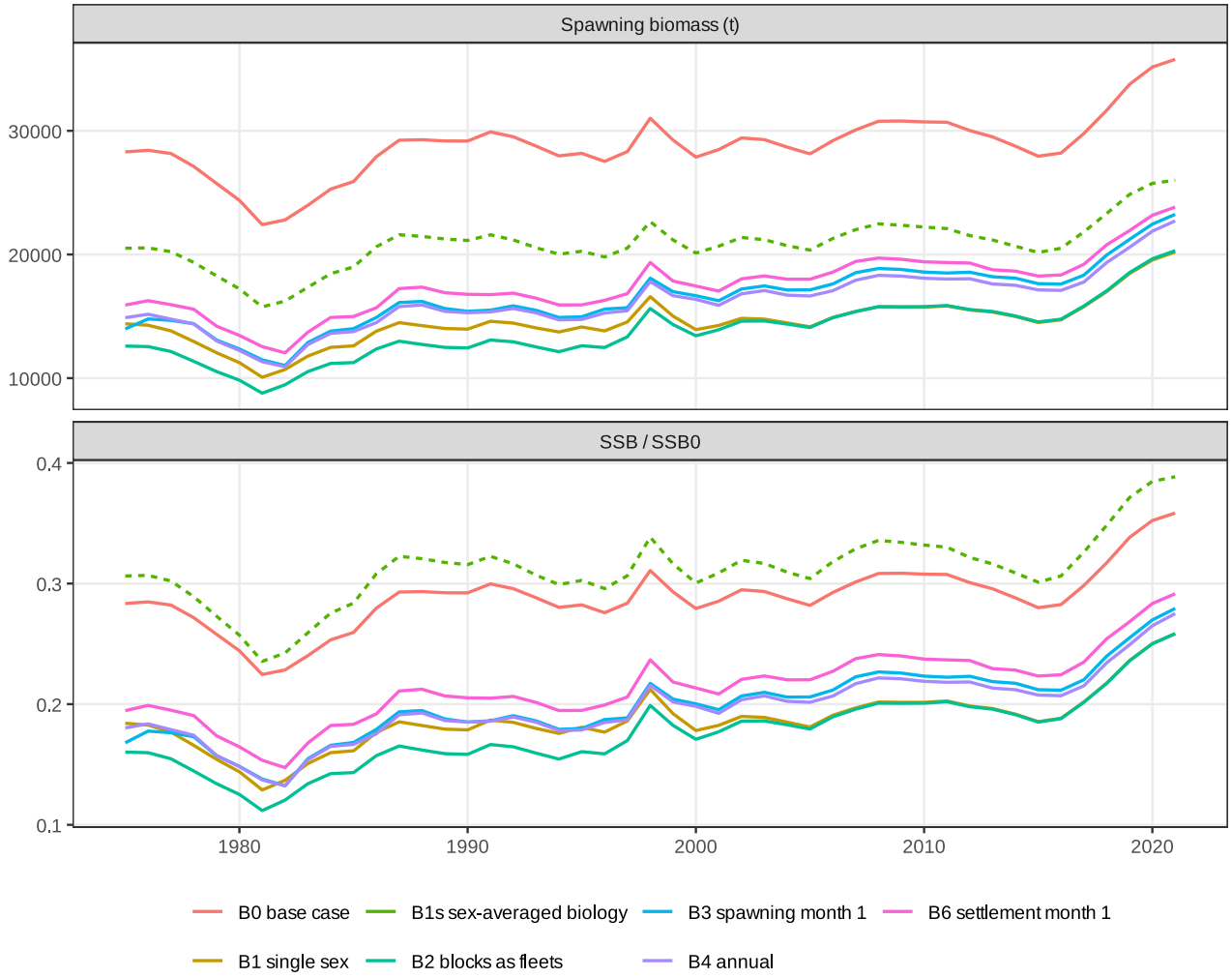


Figure 1: Female spawning biomass (top) and depletion (bottom) along the SS3 bridge. B5 and B7 coincide with B4 and B6.

Single sex (B1). This step changes the stock most. The compositions are sex-combined, so no data are lost and the length NLL changes little (481.3 to 486.0). The population dynamics change, however. Applied to every fish, the female M of 0.22 from age 4 gives males a much longer life than their own M of 0.36. Female growth also makes the males larger. SSB_0 falls from 99,808 t to 78,149 t and terminal depletion from 0.358 to 0.258. The initial F of F6 rises from 0.99 to 1.39, so the 1975 stock starts further down. B1s, with M and growth averaged over the sexes, gives SSB_0 66,937 t, depletion 0.389 and initial F 1.00. Its NLL (292.09) is lower than B1′s. The difference between B1 and B1s is the cost of choosing one sex's biology for both. In B1s, F4′s logistic inflection lies within SS3′s status margin of its upper bound of 150 cm (148.8 cm; the base case's is 148.3 cm).

Blocks as fleets (B2). At B1′s estimates B2 reproduces B1′s likelihood components exactly (checked by evaluating B2 at B1′s parameters). The only change is S2′s catchability: S2 in 1994-97 is an index of its own with its own floated q. SS3′s q mirror (link type 2) would share S2′s q. That version reproduces B1 at B1′s estimates, but the gradient of the shared q is wrong there (4.3 at B1′s optimum), and the minimiser stops short with gradients of 5-60. With its own q, B2 converges at NLL 292.653 (B1: 293.399). The stock barely moves (depletion 0.259).

Timing and seasons (B3-B7). SS3 does not allow spawning after the start of the season in which recruits settle. A one-season model with settlement in month 7 therefore needs spawning at the start of the year, which B3 introduces while the model is still quarterly. Spawning in month 1 rather than 5 removes four months of mortality before spawning. SSB_0 rises to 83,206 t and depletion to 0.279. Merging the quarters (B4) changes the timing of the catch, which is now removed through the year rather than quarter by quarter. It also

changes the fishery compositions' expected lengths, which become catch-weighted over the year. Depletion moves to 0.275, but the initial F falls from 1.59 to 0.34: in an annual model the initial equilibrium needs a much smaller F to give the same 1975 stock. Combining the quarterly compositions (B5) leaves every estimate unchanged. Within a year they share one expected composition, so the multinomial kernel is unchanged. SS3′s NLL falls from 291.7 to 42.8 only because its composition likelihood includes the observed proportions' own entropy, which differs between quarterly and annual vectors. Settlement at the start of the year (B6) makes age-0 fish a full year old at the end of the year, as in opal. Depletion moves to 0.292 and F_{MSY} from 0.180 to 0.224. Observing the indices in month 1 (B7) changes nothing for the fitted indices. With two subseasons, SS3 places observations in months 1-6 at the same time, and the indices moved are observed in months 1 or 4 (S6, a recruitment index, does not depend on timing). Only S5, with lambda 0, moves from month 7.

Starting values. The control file's INIT values are the supplied estimates, but the recruitment deviations in the control file start at zero. Estimated from the control file, the base case converges to a worse minimum (NLL 332.18 against 289.30). Each step therefore starts from the previous step's par file where the parameter layout allows (B2-B7). B1 and B1s start from the control file plus the previous step's deviations, read through the control file's `read_recdevs` option. Each step is then re-run from its own par file until its maximum gradient is below 1e-3, because the control file's INIT values are rounded to six digits. B4 starts from B3′s estimates with the initial F at 0.6 and $\ln(R_0)$ raised by 0.5. From B3′s initial F of 1.59 the annual model's initial stock is a few percent of SSB_0 and cannot support the catch.

4 Converting B7 to opal

`scripts/02-convert-fit-npo-swo-opal.r` converts B7 with `config/npo-swo-b7.yaml`. opal has 47 annual time steps of one season and ages 0-15. Recruits enter at age 0 at the start of the year from that year's spawning biomass, as in B7. The 25 fishing fleets become opal fisheries 1-25 in SS3 order, and the seven indices with lambda 1 become fisheries 26-32. The biology (weight, maturity, spawning output and M at age) is SS3′s start-of-year endgrowth table. Spawning output is mature female biomass times the female fraction 0.5.

4.1 Reconciliation

Table 4: Reconciliation with SS3 (exact comparisons). The 25 catch totals are summarised in one row.

Quantity	Max. rel. diff.	Pass
catch totals, 25 fleets	0.0e+00	TRUE
CPUE observation count (kept SS3 index fleets)	0.0e+00	TRUE
LF total input sample size	0.0e+00	TRUE
LF observation rows	0.0e+00	TRUE
unfished spawning potential per recruit	5.9e-07	TRUE
total weight summed over ages (Wt_Beg)	0.0e+00	TRUE

4.2 Growth and length-at-age

opal's growth curve does not enter the biology on this route. It only maps length-based selectivity to age and predicts the length compositions through opal's probability of length at age. SS3 builds a fishery's selectivity-at-age and its expected compositions from mid-year lengths, because the fishery's catch is taken through the year. opal's growth is therefore fitted to SS3′s mid-year length-at-age. The start-of-year lengths cannot be fitted in any case. SS3 grows age-0 fish linearly from 10 cm at settlement to L1 at age 0.5. A Schnute curve through them misses by up to 42 cm (RMSE 11.7 cm). At mid-year, the RMSE is 0.45 cm (Figure 2).

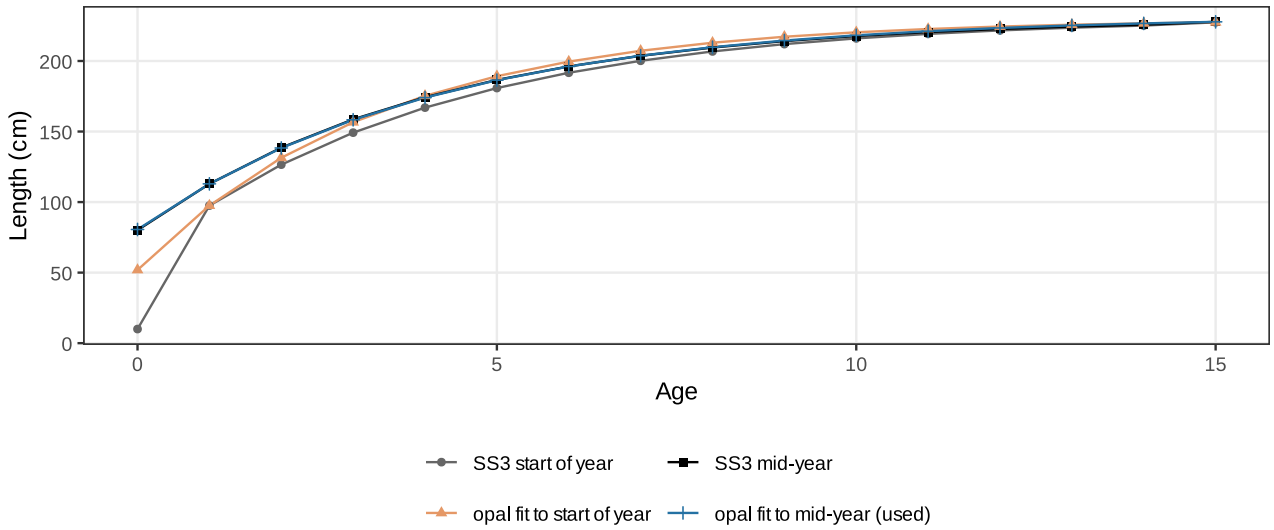


Figure 2: Length-at-age: SS3 B7 at the start of the year and at mid-year, opal's Schnute curve fitted to each (the mid-year fit is used).

4.3 Selectivity and the recruitment index

opal's double normal and logistic curves reproduce SS3's selectivity-at-length for every fishery to within 1.5e-05. The fleets that mirror another take their source fleet's parameters. Mapped to age through opal's growth, the starting selectivity-at-age is within 0.018 of SS3's for every fishery.

S6 has no opal equivalent. opal predicts an index from numbers- or biomass-at-age times a selectivity. S6 is therefore converted as an index in numbers whose selectivity is 1 at age 0 and 0 at older ages. Its observation at the start of year y then predicts that year's age-0 recruits, as SS3's does. In O0 and O1 this is a row of SS3's selectivity-at-age. `sel_fa_external` replaces the selectivity of every fishery, so in O2 (opal's own curves) S6 is an age-based (`sel_basis_f = 2`) double normal. Its peak is at age −1, so the plateau, which is at least one age wide, ends at age 0, and both limbs are steep. It gives selectivity 1 at age 0 and below 1e-4 at older ages, and is fixed. The converter reads SS3's units code 33 as biomass and S6's size pattern 0 as selectivity 1 at every age. The script overrides both.

5 opal at SS3's values

O0 is opal at B7's parameter values with SS3's selectivity-at-age, and nothing estimated.

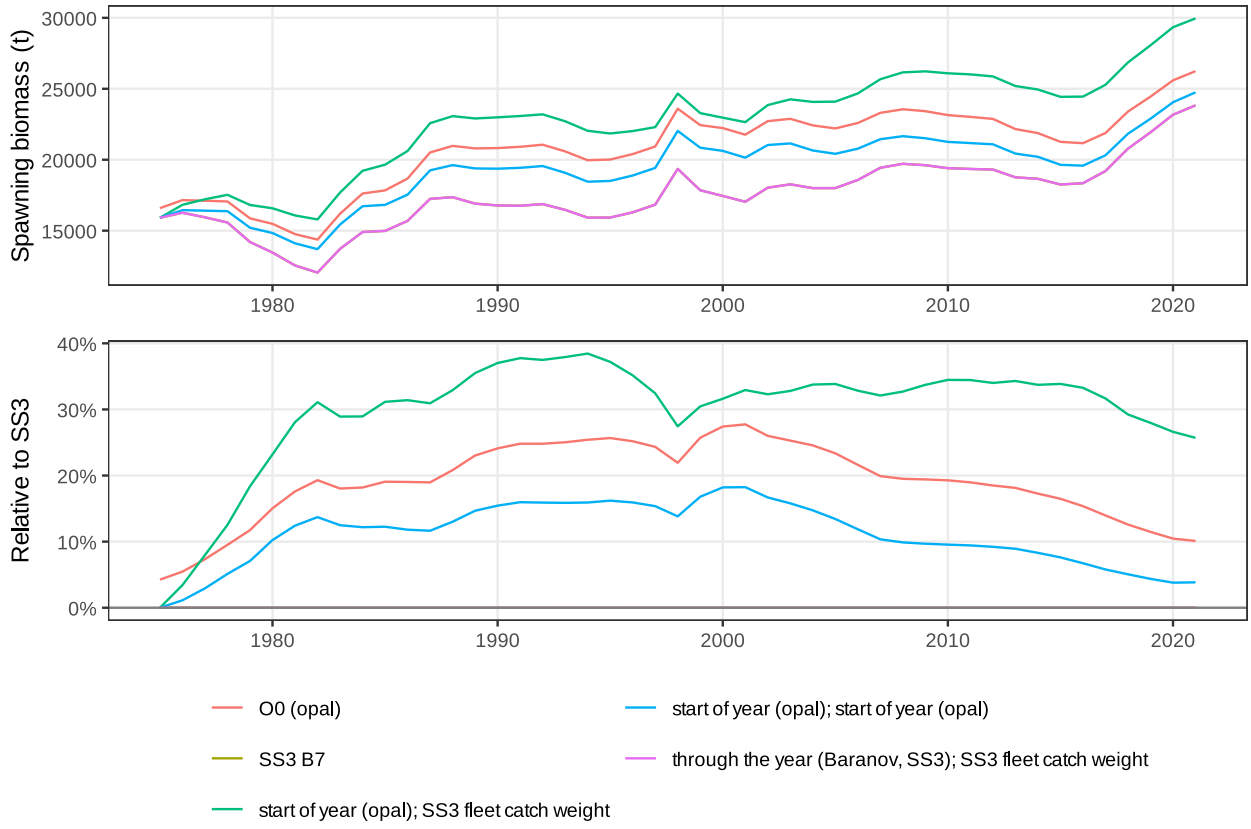


Figure 3: O0 against SS3 B7: spawning biomass (top) and relative difference (bottom). The replays start from SS3′s 1975 numbers-at-age with SS3′s recruitment each year and differ only in how the catch is removed.

Two things separate O0 from SS3.

The initial equilibrium. SS3 solves the 1975 equilibrium with F6′s initial F applied through the year. opal removes the equilibrium catch at the start of the year. opal's 1975 numbers-at-age are 0.6% above SS3′s at age 0, rising to 19% in the plus group, and its 1975 spawning biomass is 4.2% above SS3′s.

Catch timing and catch weight. Table 5 replays SS3′s dynamics (SS3′s 1975 numbers and SS3′s recruits) with each year's catch removed in different ways. Removing it through the year (Baranov, as SS3′s hybrid F), with each fleet's SS3 catch weight-at-age, reproduces SS3 to 4.0e-06. This confirms that the converted catch, selectivity, M, spawning output and units are SS3′s. Removing the catch at the start of the year, as opal does, leaves spawning biomass up to 18% above SS3′s. The start-of-year weights opal uses partly offset this. With SS3′s (larger) catch weights, a start-of-year removal takes fewer fish of the weight-based fleets, and the difference grows to 38%. A mid-year (Pope) removal is within 5%.

Table 5: SS3′s dynamics replayed with different catch removals: spawning biomass relative to SS3 B7 in 2021 and the largest absolute difference over 1975-2021.

Removal	Catch weight	2021	Max.
start of year (opal)	start of year (opal)	3.8%	18.2%
start of year (opal)	SS3 fleet catch weight	25.7%	38.5%
mid-year (Pope)	SS3 fleet catch weight	−2.1%	4.9%
through the year (Baranov, SS3)	mid-year	−4.9%	5.3%
through the year (Baranov, SS3)	SS3 fleet catch weight	0.0%	0.0%

6 opal fits

Growth, steepness (0.9), sigma_R (0.42), M and the biology stay at SS3′s values, as in SS3. opal's default priors are switched off (SS3′s prior lambda is 0) and the length compositions carry SS3′s lambda of 0.5. Every fit estimates B_0, the recruitment deviations for 1976-2021, the initial-age deviations of the cohorts born 1965-1975, F6′s initial F and the seven catchabilities. SS3 floats q, and opal estimates it.

- **O1**: SS3′s selectivity-at-age, fixed.
- **O2**: opal's selectivity curves, started at SS3′s. The parameters SS3 estimates are estimated (26, with the mirrors sharing their source fleet's).
- **O2b**: O2 refitted with the selectivity parameters it cannot determine fixed at their O2 values (Table 7).
- **O1-noLF**: O1 without the length compositions.
- **O1-noS6**: O1 without the recruitment index (the report's sensitivity 15).

Table 6: Fits. The objective values are opal's and are not comparable with SS3′s likelihood (different constants).

Model	Est. par.	Objective	Max. grad.	PD Hessian	B0	Init. F (F6)	SSB2021	SSB2021/B0
SS3 B7	86				81,740	0.326	23,833	0.292
O0	0	5,747.70			81,740	0.326	26,243	0.321
O1	66	2,493.04	5.7e-12	TRUE	82,238	0.347	27,607	0.336
O2	92	2,442.69	1.6e+00	FALSE	80,714	0.474	24,594	0.305
O2b	84	2,442.69	9.1e-12	TRUE	80,714	0.474	24,594	0.305
O1-noLF	66	−47.87	2.5e-09	TRUE	80,572	0.328	24,119	0.299
O1-noS6	65	2,505.42	1.1e-11	TRUE	83,504	0.360	28,730	0.344

O1 converges with a positive-definite Hessian. It estimates a stock of similar size (B_0 82,238 against 81,740 t) that is less depleted than SS3′s (0.336 against 0.292 in 2021). This is in the direction and of roughly the size of O0′s offset, which comes from catch timing and the initial equilibrium. Without the compositions (O1-noLF) depletion is 0.299. Without S6 (O1-noS6) it is 0.344, so the recruitment index has little leverage on opal's estimate of the stock's state.

O2 reaches a lower objective than O1 (2,442.7 against 2,493.0), but stops with a gradient of 1.6e+00 and a Hessian that is not positive definite. Four of its selectivity parameters end on SS3′s bounds. F2′s logistic inflection (SS3′s bound of 150 cm; SS3′s own estimate for F4 is also close to its 150 cm bound) is one. The descending widths of F3, F6 and F8 reach SS3′s lower bounds, where the likelihood is flat because their plateaus extend to the largest length. Fixing these, and then the parameters that dominate the remaining flat directions of the Hessian (Table 7), gives O2b. O2b has the same objective, converges with a positive-definite Hessian, and estimates terminal depletion 0.305.

Table 7: Selectivity parameters fixed in O2b (opal scale), and why: on a bound in O2, or dominating a flat direction of the Hessian (eigenvalue below 1e-6). Columns of the double normal: 1 peak, 2 top, 3 ascending width, 4 descending width; of the logistic: 1 inflection, 2 width.

Fishery	Name	Column	Value	Reason
2	F2_TWN_WCNPO_DWLL_late	1	0.17	bound
3	F3_US_WCNPO_LL_shallow_late	4	−14.62	bound
6	F6_JPN_WCNPO_OSDWLL_early_Area1	4	−10.62	bound
8	F8_US_WCNPO_LL_shallow_early	4	−14.62	bound
7	F7_JPN_WCNPO_CODF	2	2.37	flat
7	F7_JPN_WCNPO_CODF	4	−4.23	flat
20	F1_JPN_WCNPO_OSDWCOLL_late_Area1_1994_97	4	8.04	flat
8	F8_US_WCNPO_LL_shallow_early	2	−7.86	flat



Figure 4: Spawning biomass, depletion and age-0 recruitment: SS3 B7 and the opal models (O2, not converged, is not shown; O2b has the same estimates).

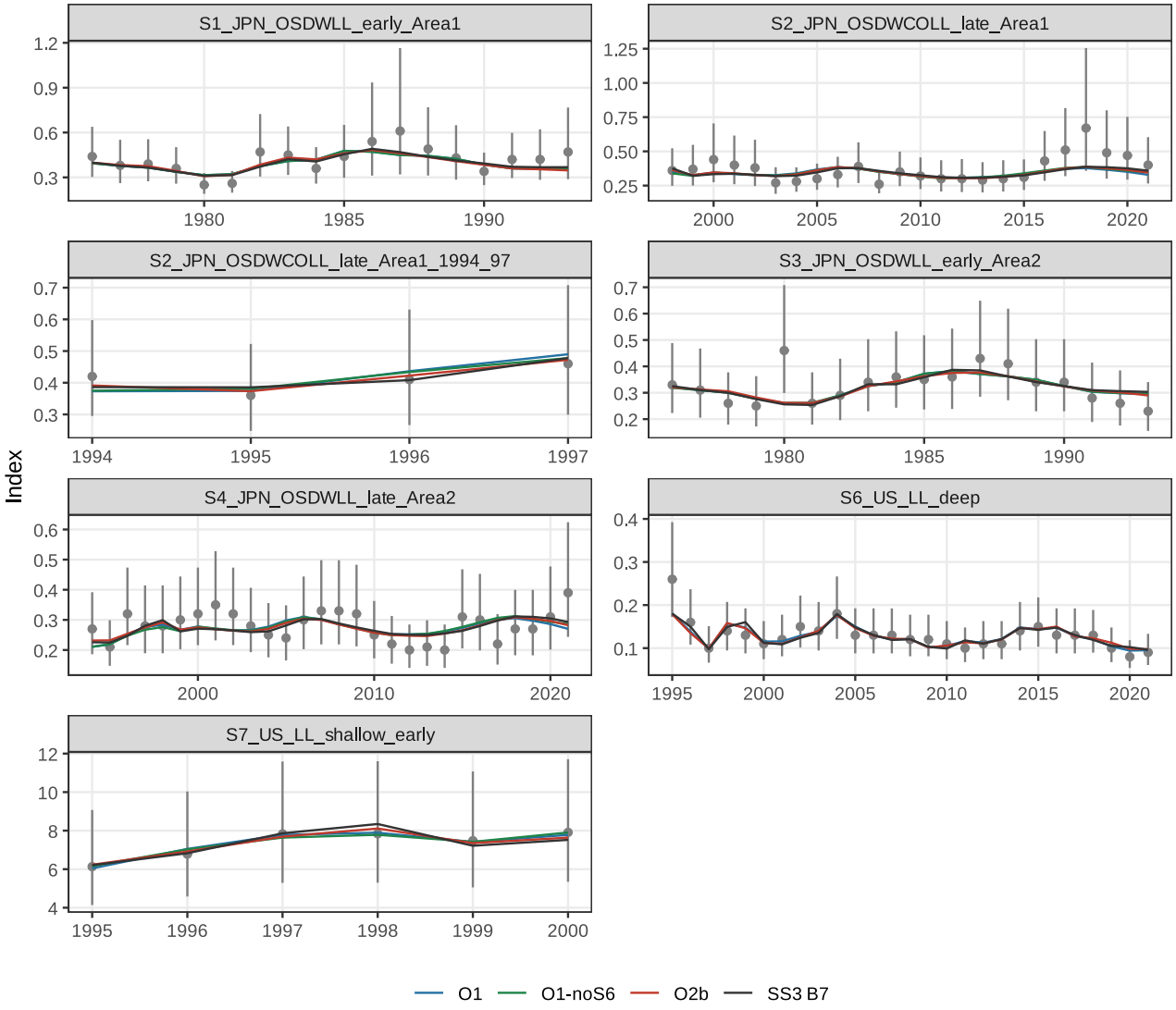


Figure 5: Fits to the indices (points: observed, with 95% intervals from the input SE). S6 is the recruitment index; O1-noS6 does not fit it (not shown for S6).

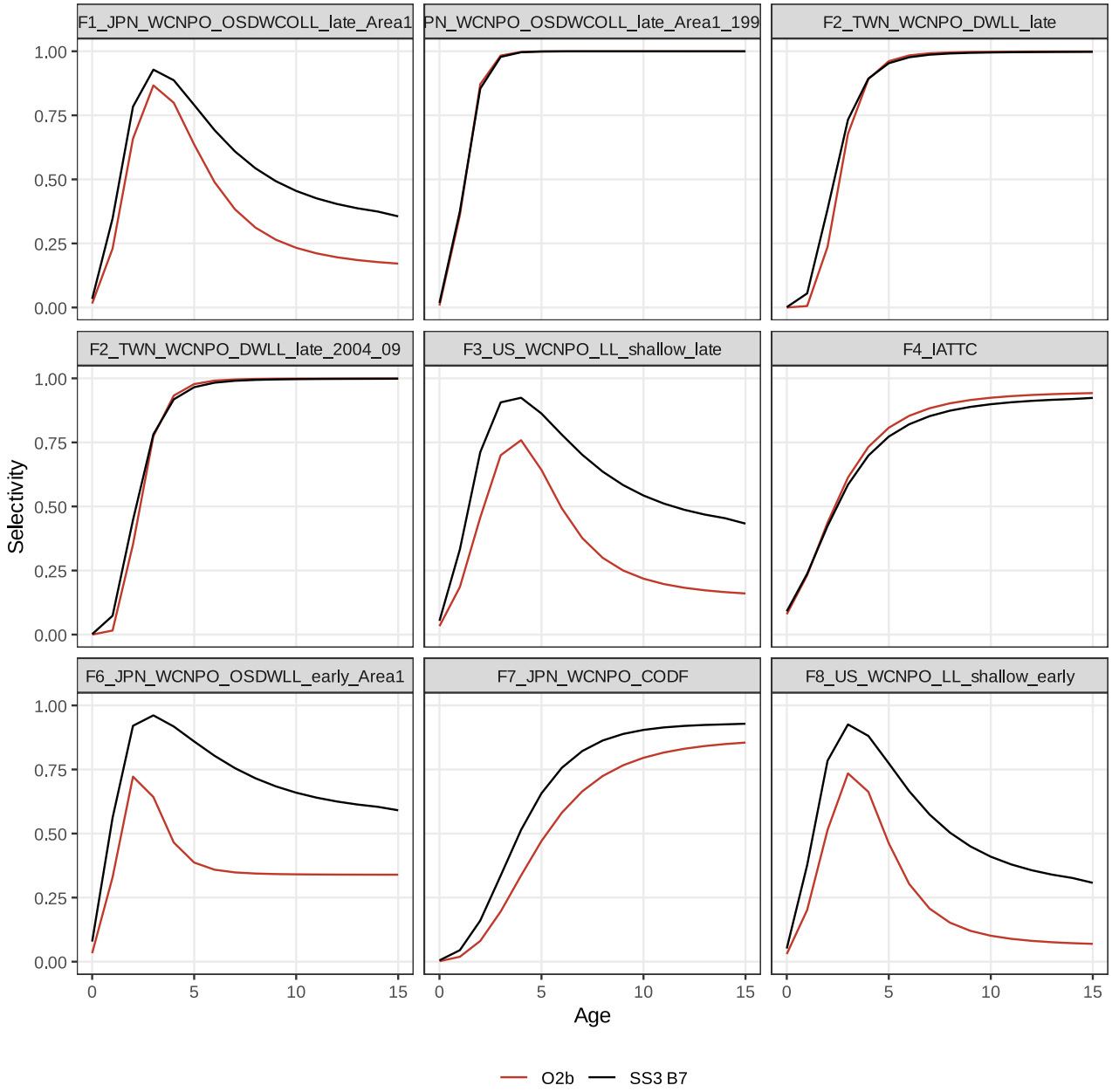


Figure 6: Selectivity-at-age in 2021: SS3 B7 (black) and opal O2b (red), for the fisheries with length compositions and the source fleets of the indices.

7 Limitations and differences from SS3

7.1 Not available in opal

- **Two sexes.** opal tracks one population vector and scales spawning potential by a sex ratio. Modelling the base case needs a sex dimension in the numbers-at-age, sex-specific growth and M, and spawning biomass from females only. The compositions here are sex-combined, so sex-specific composition likelihoods are not needed. The predicted combined composition would sum the two sexes' catches. B1 against B1s and B0 shows that the choice of single-sex biology moves terminal depletion between 0.258 (female) and 0.389 (sex-averaged), against 0.358 with two sexes.
- **Seasons with annual ages.** opal has `n_season`, and within a season its catch and index predictions are seasonal. However:
  - it predicts compositions from the year's total catch-at-age with one length-at-age for the whole year, not by season;
  - it has one weight-at-age vector for the year;

- it spawns and recruits only at the start of season 1, whereas the base case spawns in month 5 and recruits settle in month 7.

The SS3-to-opal converter handles one season per year only. Seasonal opal models would also need season-specific length-at-age (growth evaluated at the season's midpoint) to predict quarterly compositions.

- **Spawning and settlement timing.** opal's spawning biomass is that at the start of the year, before any mortality. B3 shows that moving spawning from month 5 to month 1 raises SSB_0 by 6% and terminal depletion from 0.259 to 0.279. A spawning-time fraction (M and F applied before spawning) would remove this step.
- **Time-varying selectivity.** Blocks cannot be estimated. Blocks that replace parameters are reproduced exactly by splitting the fleet (B2), but every fleet that mirrors a blocked fleet has to be split too (here seven fleets). A split index needs its own catchability or a shared one, and SS3′s shared q (link 2) does not converge.
- **Recruitment index.** opal has no index of the year's recruitment (SS3 units 33). An index in numbers with age-0 selectivity reproduces it at the start of the year. Because `sel_fa_external` replaces every fishery's selectivity, a fit with opal's own curves needs an age-based curve pinned to that shape. A per-fishery external selectivity, or an index type for recruitment (or for any age range), would avoid the workaround.
- **Analytical (floated) catchability.** SS3 solves q; opal estimates it.

7.2 Structural differences handled in the conversion

- **Catch timing.** opal removes each season's catch at its start; SS3′s hybrid F removes it through the season. At SS3′s values this leaves spawning biomass up to 18% above SS3′s (Table 5). A mid-season (Pope) removal would be within 5%. Quarterly opal seasons would also approximate SS3′s removal.
- **Initial equilibrium.** opal's initial equilibrium removes the initial-F catch at the start of the year. Its 1975 spawning biomass at SS3′s values is 4.2% above SS3′s. A Baranov or mid-year removal in `get_initial_numbers()` would match SS3.
- **Catch weight.** opal converts catch in weight to numbers with one weight-at-age (start of year here). SS3 uses each fleet's catch weight-at-age (selectivity-weighted, mid-season).
- **Length-at-age for selectivity and compositions.** SS3 forms a fishery's selectivity-at-age and expected compositions from mid-year lengths. opal uses one length-at-age, here fitted to SS3′s mid-year lengths. The biology stays at the start of the year.
- **Growth below the first reference age.** SS3 grows age-0 fish linearly from settlement to L1 at age 0.5. opal's Schnute curve cannot represent this, which matters only at the start of the year.
- **Fleet layout.** SS3 3.30 allows fishing fleets after the surveys (here the block fleets 28-34). The converter maps SS3 fleets to opal fisheries by fleet type.

7.3 What the bridge itself changes

The single-sex step is not neutral. B1 applies female M and growth to every fish. Males live longer and grow larger than they do in the base case, and the stock is smaller and more depleted. B1s, with sex-averaged biology, is closer to the base case. Either choice is an approximation that opal's lack of sexes forces. With the sex-combined compositions here, a two-sex opal would need no new data. The timing steps (B3, B6) move spawning and settlement to the start of the year. B3 is required for SS3 itself to merge the seasons, and both change the stock's scale and productivity (SSB_{MSY}, F_{MSY}). These are differences between the base case and opal's structure, not between SS3 and opal.

Reproducing this document

Step	Script	Writes
1	`scripts/01-build-ss3-bridge.r`	the SS3 bridge models and `vignettes/data/npo-swo-ss3-bridge-cache.rds`

Step	Script	Writes
2	`scripts/02-convert-fit-npo-swo-opal.r`	`vignettes/data/npo-swo-opal-cache.rds` and the fitted models
3	`quarto render vignettes/npo-swo-ss3-to-opal.qmd`	this document

```
Caches: bridge 2026-11-13 17:33:29 UTC (SS3 3.30.22.1); opal 2026-11-13 17:35:10 UTC (opal
0.1.3, r4ss 1.56.0, R 4.5.3).
```

Bibliography

[1] ISC Billfish Working Group, "Stock Assessment Report for Swordfish (Xiphias gladius) in the North Pacific through 2021," Kanazawa, Japan, Report of the 23rd Meeting of the ISC, Annex 11 ISC/23/ANNEX/11, Jul. 2023.

[2] R. D. Methot and C. R. Wetzel, "Stock Synthesis: A biological and statistical framework for fish stock assessment and fishery management," *Fisheries Research*, vol. 142, pp. 86–99, 2013, doi: 10.1016/j.fishres.2012.10.012.